REAL-TIME INTRUSION DETECTION SYSTEM

Machine Learning Network Security Report

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1. Executive Summary

This report summarizes the network traffic analyzed by the Real-Time Intrusion Detection System. The system uses machine-learning-based classification to identify network flows as either normal traffic or potential intrusions.

Metric	Value
Total Network Flows	2711
Normal Flows	2547
Detected Attacks	164
Attack Rate	6.05%

2. Machine Learning Model

Parameter	Value
Algorithm	Random Forest Classifier
Training Features	duration, protocol_type, src_bytes, dst_bytes, count
Classification	Normal / Attack
Optimized Threshold	0.30
Validation Accuracy	80.96%
Attack Precision	0.93
Attack Recall	0.72
Attack F1-Score	0.81

3. Network Protocol Distribution

Protocol	Flow Count
TCP	1651
UDP	1058
OTHER	2

4. Detected Attack History

Time	Source	Destination	Protocol	Confidence
2026-08-25 17:02:46	10.2.0.1:65432	192.168.1.18:57957	TCP	42.68%
2026-08-25 17:01:40	150.171.110.85:443	192.168.1.18:65035	TCP	48.45%
2026-08-25 17:01:39	10.2.0.1:65432	192.168.1.18:62778	TCP	42.68%
2026-08-25 17:01:17	192.168.1.18:64452	8.8.4.4:443	UDP	48.98%
2026-08-25 17:01:16	192.168.1.18:58934	202.88.211.77:443	UDP	31.79%
2026-08-25 17:00:31	10.2.0.1:65432	192.168.1.18:51706	TCP	50.96%
2026-08-25 17:00:20	192.168.1.18:51564	4.150.223.108:443	TCP	42.2%
2026-08-25 16:28:06	10.2.0.1:65432	192.168.1.18:55386	TCP	42.68%
2026-08-25 16:28:04	172.217.17.206:443	192.168.1.18:64258	UDP	85.5%
2026-08-25 16:28:04	192.168.1.18:5353	224.0.0.251:5353	UDP	36.71%
2026-08-25 16:28:02	10.2.0.1:65432	192.168.1.18:49874	TCP	42.68%
2026-08-25 16:27:47	192.168.1.18:58127	3.111.241.224:443	TCP	92.55%
2026-08-25 16:27:32	192.168.1.18:58127	3.111.241.224:443	TCP	35.49%
2026-08-25 16:27:30	192.168.1.18:50919	4.224.112.242:443	TCP	41.2%
2026-08-25 16:27:22	192.168.1.18:60443	23.205.101.152:443	TCP	35.83%
2026-08-25 16:26:20	192.168.1.18:49672	23.211.60.116:443	TCP	36.6%
2026-08-25 16:25:46	192.168.1.18:63265	8.8.8.8:443	TCP	88.91%
2026-08-25 16:25:43	192.168.1.18:57464	57.144.211.32:443	TCP	93.41%
2026-08-25 16:25:39	192.168.1.18:58786	8.8.4.4:443	TCP	88.91%
2026-08-25 16:25:19	192.168.1.18:63265	8.8.8.8:443	TCP	88.91%

5. Security Recommendations

1. Investigate repeated attack predictions from the same source address.

2. Monitor network flows with unusually high traffic volume or packet counts.

3. Review suspicious TCP and UDP connections identified by the ML classifier.

4. Use the attack confidence score together with network context when validating alerts.

5. Continue collecting representative network traffic to improve the IDS model.

6. Periodically retrain and evaluate the model using updated datasets.

6. Conclusion

The Real-Time Intrusion Detection System combines live network packet capture, flow-level feature extraction, machine-learning classification, threshold optimization, CSV logging and dashboard-based monitoring. The generated statistics and attack history provide a summary of the traffic observed during the monitoring period.